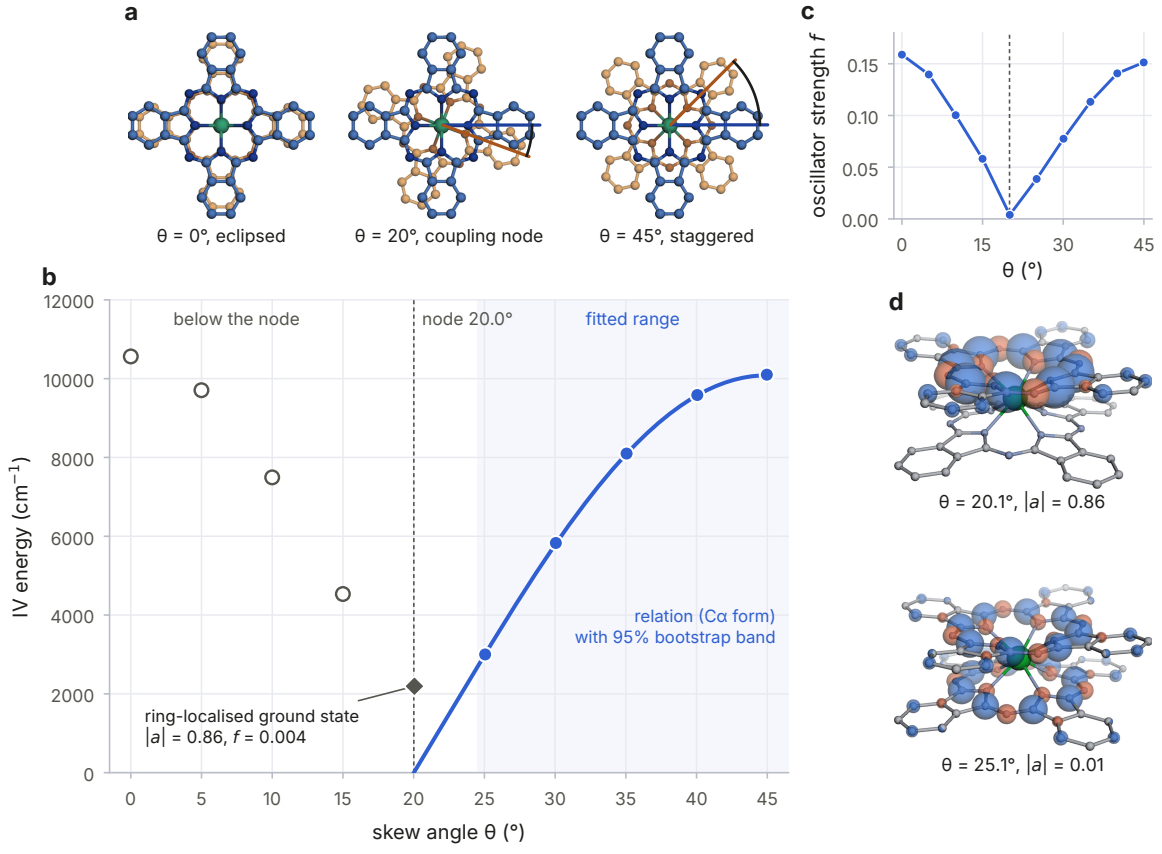

a
θ = 0°, eclipsed
θ = 20°, coupling node
θ = 45°, staggered
c
oscillator strength f
0.15
0.10
0.05
0.00
0
15
30
45
θ (°)
b
12000
10000
8000
6000
4000
2000
0
IV energy (cm⁻¹)
below the node
node 20.0°
fitted range
relation (Cα form)
with 95% bootstrap band
ring-localised ground state
|a| = 0.86, f = 0.004
0
5
10
15
20
25
30
35
40
45
skew angle θ (°)
d
θ = 20.1°, |a| = 0.86
θ = 25.1°, |a| = 0.01